AMADEUS SCHEMA INTEGRATION AT THE STORAGE BOUNDARY

Ordinary calls • recording during science • validation after calculate / during load / at saved open

AI: development, review and documentation only. Runtime: R / JSON Schema / SQL. Human approval: pending.

01 Build contracts and catalog

Build / changed release only

dev/build_dataset_metadata.R; build_parquet_contract.R; check_approval.R; validate_release.R

IN: LinkML definitions, package code/labels, unit policy and evidence → OUT: Pinned JSON Schema, SQL, catalog and runtime contract

Approve authoritative definitions and native-unit/time policy. Build consistency does not equal scientific approval.

02 User request

Before science

Ordinary caller; examples/ordinary_with_recording.R is a test example

IN: Variables, dates, locations, paths and method arguments → OUT: Actual request passed to unchanged Amadeus calls and recorder

User chooses request. No runtime spreadsheet or AI service; tested scope is explicit.

03 Amadeus download

Science + recording

amadeus::download_data; R/provenance_recorder.R observes

IN: Request and provider assets or available cache → OUT: Source files, recorded source hashes and call events

No package patch. Cache identity is not proof of a fresh provider download.

04 Amadeus process

Science + recording

amadeus::process_covariates; provenance_recorder.R observes

IN: Source files, dates and site extent → OUT: Processed raster/TIFF and recorded call/output context

Existing processing behavior retained; capture changes and environment identities in any replay.

05 Amadeus calculate

Science + recording

amadeus::calculate_covariates; provenance_recorder.R observes

IN: Processed raster, sites and method → OUT: Ordinary R result frames and sealed execution context

Scientific computation ends here. Recorder operates alongside calls; it does not enforce the schema.

06 Verify actual context

After calculation

R/export_gridmet.R; R/provenance_recorder.R verification

IN: Completed calculation outputs, actual request and recording → OUT: Verified context for export, or rejection

No pilot example_gridmet.R orchestration or schema_validator.R checks in the active path. Core Amadeus checks remain.

07 Assemble and validate records

After calculation / before publication

R/schema_export.R; R/storage_boundary.R

IN: Actual outputs/context, compiled catalog and contract → OUT: Accepted schema instance or rejection

Deterministic JSON Schema/R/SQL check structure, units/binding, dates, membership, finite values and lineage.

08 Write and read typed Parquet

Storage boundary

R/storage_boundary.R; R/parquet_experiment.R

IN: Mapped actual records and compiled storage mapping → OUT: Typed Parquet and full-column readback evidence

DuckDB provides Parquet I/O. Typing/metadata alone do not establish scientific validity or true history.

09 Load and verify DuckDB

During loading / before completion

R/database.R; R/parquet_experiment.R

IN: Accepted records, Parquet and generated DDL/integrity SQL → OUT: Loaded DB, integrity/fidelity and reopen receipts

102 SQL queries include empty tables. Schema acceptance alone is not a successful database publication.

10 Bind published artifacts

After successful load

R/storage_boundary.R: storage_publish_manifest

IN: Actual owned artifact bytes and matching release/instance → OUT: Manifest hashes for files and receipts

Unsigned local integrity, not authenticated history or protection against coordinated privileged rewriting.

11 Complete and publish pointer

Publication

R/schema_export.R

IN: Successful storage checks and bound manifest → OUT: Read-only critical files, completed workflow and final LATEST

Failed load must not complete or advance pointer. Not multi-file power-loss durability.

12 Verify open and query

Each supported saved-store open

R/database.R; storage_verify_publication

IN: Completed publication, file hashes and release identity → OUT: Verified read-only DuckDB query, or rejection

No raw-source reread for saved query. Direct DBI/IDE access bypasses this verification path.

13 Review evidence and decide

Qualification / release review

Separate tests/direct_source_check.R and human review records

IN: Source comparisons, adversarial tests, replay, known limits → OUT: Scoped human decision or pending status

Units, physical time, binding, provenance truth and acceptance need human decisions. Production approval remains pending.

Contract consistency is checked at build; actual-data checks follow calculate. Schema acceptance alone is not database publication. Gap closure requires tests plus human decisions.